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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations September-2023

Computer Science and Engineering

PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART A

1	1.1	Differentiate between <i>SIMD</i> and <i>MIMD</i> .	02
	1.2	Suppose you want to achieve a speed up of 80 with 100 processors. What fraction of the original computation can be sequential?	02
	1.3	_____ is used to handle if statements in vector loops.	01
	1.4	Mention the use of Gather-scatter with respect to vector architecture.	01
	1.5	Define Chime.	01
	1.6	Differentiate between vector Architecture and <i>GPU</i> .	02
	1.7	Use the <i>GCD</i> test to determine whether dependency exist in the loop $for(i = 0; i < 100; i++)$ $\{ X[2 * i + 3] = X[2 * j] * 5.0; \}$	02
	1.8	Define Tasks and decomposition.	01
	1.9	Define critical path.	02
	1.10	Differentiate between openMP and <i>MPI</i> .	02
	1.11	Mention any four characteristics of tasks.	02
	1.12	Give the syntax of <i>MPI</i> -gather.	02

PART B

2	a	Differentiate between centralized shared memory and distributed memory.	05
	b	Illustrate snooping protocol with an example.	05
	c	Illustrate cache coherence protocol for write back cache with the help of state transition diagram.	06
3	a	Illustrate with a neat diagram the basic structure of a vector architecture.	08
	b	Show the code for <i>MIPS</i> and <i>VMIPS</i> for <i>DAXPY</i> loop. Assume that the starting address of <i>X</i> and <i>Y</i> are in <i>Rx</i> and <i>Ry</i> respectively.	08
		OR	
4	a	Using multiple functional units how do you improve the performance of single vector add instruction $C = A + B$.	08
	b	Briefly describe about scheduling of threads of <i>SIMD</i> instruction.	08

5	a	Illustrate the different decomposition techniques with example.	10
	b	Briefly describe the methods for containing interaction overhead.	06
OR			
6	a	Illustrate all the parallel algorithm models.	10
	b	Write a iterative program for finding minimum element in an array of numbers A of length n .	06
7	a	Illustrate the different message passing interface.	10
	b	Give the syntax of Barrier, Broadcast and Reduction which is used for collective communication and computation operations in <i>MPI</i> .	06
8	a	Illustrate three steps to accelerate with openACC.	06
	b	Illustrate the openACC directives in detail.	10